

Wi-Fi Series

Data Sheet

USE_8720_CF

Wi-Fi+BLE Module

Version

VERSION	REVISION DATE	REVISOR	REVIEWER	REVISED CONTENTS
1.0	03/05/2025	AD	AK	BLE/Wi-Fi Module
2.0	04/08/2025	DJ	AD	Added Assembly recommendation and Module placement strategy
3.0	22/09/2025	AD	MB	Added storage temperature
4.0	23/09/2025	AD	DJ	Modified BLE-Wifi to Wifi+Ble & BLE series to Wi-Fi series

Table of Contents

SI NO	CONTENTS	PAGE NO
1	Features	03
2	General Description	04
3	Pin Definition	04
4	Pin descriptions	05
5	Power Supply DC Characteristics	06
6	Digital IO Pin DC Characteristics	07
7	Module Diagram	08
8	Module Dimension Drawing	09
9	Recommended Footprint Dimension	10
10	Recommended Footprint Clearance	10
11	Recommended Connections	11
12	Recommended Assembly	12
13	Module placement strategy	13

Features:

General

- Bluetooth Low Energy (BLE) 4.2
- Supports LE secure connections
- Supports LE scatternet
- Supports 1 Master/slave
- Operating temperature: -40 to +105 °C

MCU Platform

- Real-M300 (KM4) clock frequency up to 100MHz
- I-Cache 32KB/D-Cache 16KB
- Supports DMA
- eXecute In Place (XIP) on flash

Wi-Fi features

- 802.11 b/g/n compatible 1x1, 2.4GHz
- 802.11e QoS Enhancement (WMM)
- Wi-Fi WEP,WPA,WPA2,WPA3,WPS. Open, shared key, and pair-wise key authentication services
- Supports low power Tx/Rx for short-range application
- Supports Antenna diversity
- Frame aggregation for increased MAC efficiency (A-MSDU, A-MPDU)
- Low latency immediate High- Throughout Block A Acknowledgment (HT-BA)
- Long NAV for media reservation with CF-End for NAV release
- Integrated balun,PA/LNA

Peripheral Interfaces

- 3 x UART interface, baud rate up to 4MHz and all of them can be configurable as log UART
- 1 x I2C, Max clock 400Kbps
- 1 x SDIO 2.0 Device, up to 50MHz
- 1 x SPI, Master clock up to 25Mbps/Slave clock up to 5Mbps
- 8 x PWM with configurable duration and duty cycle from 0~100%
- 1 GDMA with 2 channels

Secure

- Supports secure boot
- Crypto engine: MD5,SHA-1,SHA2-224,SHA2-256,HMAC,AEC

General Description:

The USE_8720_CF is based on highly integrated single-chips with a low-power IEEE 802.11n Wireless LAN(WLAN) compatible network controller. This uses a Real-M300(KM4) CPU that is based on ARMv8 -M architecture, and integrates a WLAN MAC,an 1T1R capable WLAN baseband, an RF circuit, and Bluetooth Low Energy (BLE). They also provide configurable GPIOs that are configured as digital peripherals for various applications and control usage.

For more information reach out to us at info@unisemindia.com.

Pin Definition:

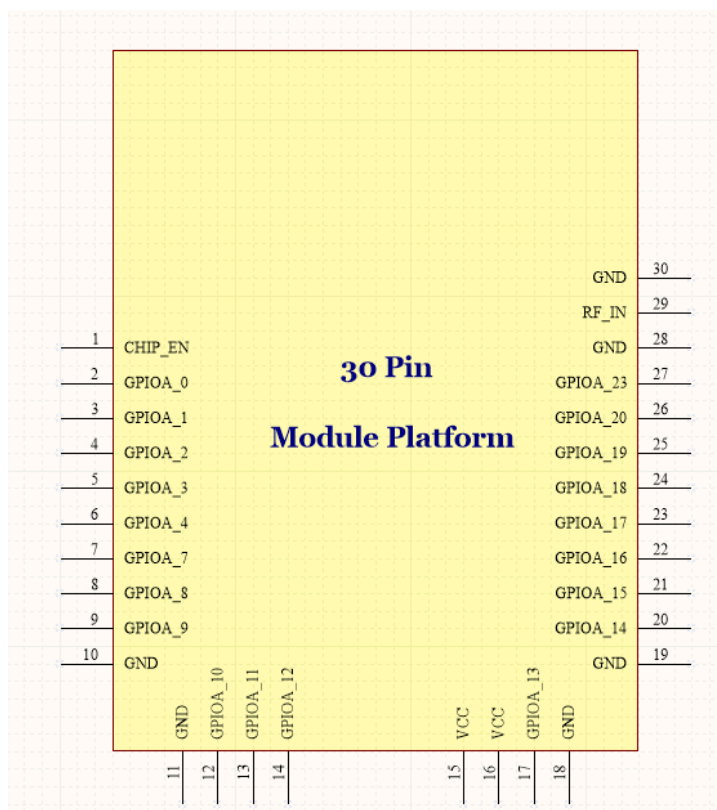


Fig 1 : Module Pinouts

Pin Descriptions:

Pin No.	Pin Name	Type	Function Description
1	CHIP_EN	I	1: Enable chip 0: Shutdown chip
2	GPIOA_0	I	1: Enter into test/debug mode 0: Normal operation mode (GPIOA_0, GPIOA_13)=(1,1): Enter into download image mode GPIO pin
3	GPIOA_1	I	1: eFUSE settings are not loaded 0: eFUSE settings are loaded GPIO pin
4	GPIOA_2	IO	GPIO pin
5	GPIOA_3	IO	GPIO pin
6	GPIOA_4	IO	GPIO pin
7	GPIOA_7	IO	GPIO pin
8	GPIOA_8	IO	GPIO pin
9	GPIOA_9	IO	GPIO pin
10	GND	GND	Ground pin
11	GND	GND	Ground pin
12	GPIOA_10	IO	GPIO pin
13	GPIOA_11	IO	GPIO pin
14	GPIOA_12	IO	GPIO pin
15	VCC	PI	3.3V power input
16	VCC	PI	3.3V power input
17	GPIOA_13	IO	GPIO pin

18	GND	GND	Ground pin
19	GND	GND	Ground pin
20	GPIOA_14	IO	GPIO pin
21	GPIOA_15	IO	UART2_IN
22	GPIOA_16	IO	UART2_OUT
23	GPIOA_17	IO	GPIO pin
24	GPIOA_18	IO	GPIO pin
25	GPIOA_19	IO	GPIO pin
26	GPIOA_20	IO	GPIO pin
27	GPIOA_23	IO	1: LDO 0:SWR GPIO pin
28	GND	GND	Ground pin
29	RF_IN	IO	WL RF signal
30	GND	GND	Ground pin

The signal type codes are used in the table:

I:Input; O:Output; PI:Power input; PO:Power output

Power Supply DC Characteristics:

Parameter	Symbol	Min	Typ.	Max	Units
DC supply voltage for VDD_IN (3.3V)	VDD_IN	2.97	3.3	3.63	V
DC supply voltage for 3.3V power rail	VD33_SDIO VA33_XTAL VA33_SYN VA33_RF VA33_PA	2.97	3.3	3.63	V

	VA33_TR VD33_FLASH VD33_OUT				
DC supply voltage for 1.1V power rail	VD11_CORE	1.09	1.146	1.20	V
DC supply voltage for GPIO/Embedded flash	VD1833_LPC	2.97	3.3	3.63	V
DC supply voltage for Embedded PSRAM1	VD1833_LPC	1.7	1.8	1.95	V

Digital IO Pin DC Characteristics:

Symbol	Parameter	Conditions	Min	Typ.	Max	Units
V _{IH}	Input-High voltage	LVTTL	2.0	-	-	V
V _{IL}	Input-low voltage	LVTTL	-	-	0.8	V
V _{OH}	Output-High voltage	LVTTL	2.4	-	-	V
V _{OL}	Output-low voltage	LVTTL	-	-	0.4	V
V _{T+}	Schmitt-Trigger High level	-	1.377	1.683	1.908	V
V _{T-}	Schmitt-Trigger low level	-	0.729	0.957	1.116	V
I _{IL}	Input-Leakage current	V _{IN} =3.3V or 0	-10	± 1	10	μA
-	Driving for normal pins	-	4	-	16	mA
I _{OH}	Driving for 5V UART pins(GPIOA_13/GPIOA_14)	3.3V/5V	4/4	-	8/8	mA
I _{OL}	Driving for 5V UART pins(GPIOA_13/GPIOA_14)	3.3V/5V	4/4	-	8/8	mA
-	Driving for SDIO Device pins	-	4	-	16	mA
-	Loading for normal pins	-	-	15	-	pF

-	Loading for 5V UART pins (GPIOA_13/GPIOA_14)		-	15	-	pF
-	Loading for SDIO Device pins	-	-	15	-	pF
-	Pull resistance for normal pins`	3.3V	-	75	-	kΩ
-	Pull resistance for 5V UART pins(GPIOA_13/GPIOA_14)	3.3V/5V	-	80/120	-	kΩ
-	Pull resistance for SDIO Device pins	3.3V	-	50	-	kΩ

Module Diagram:

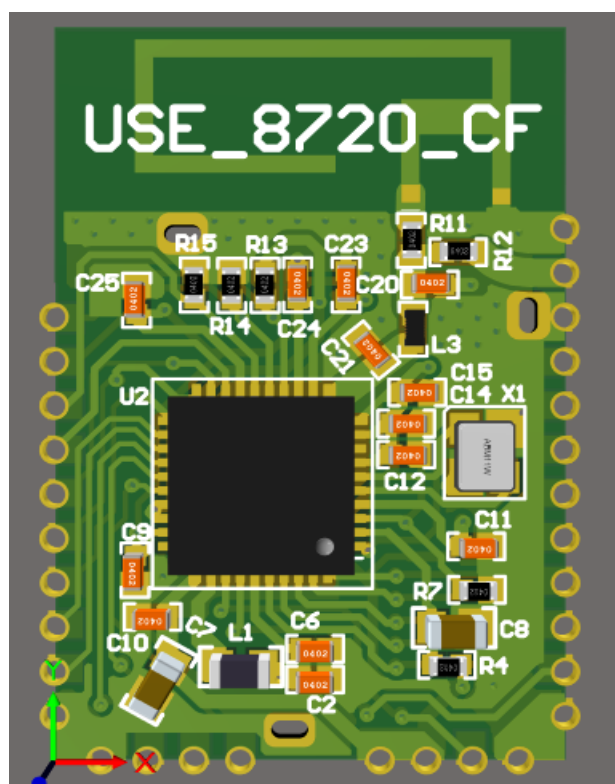


Fig 2 : Module without shield

Module Dimension Drawing:

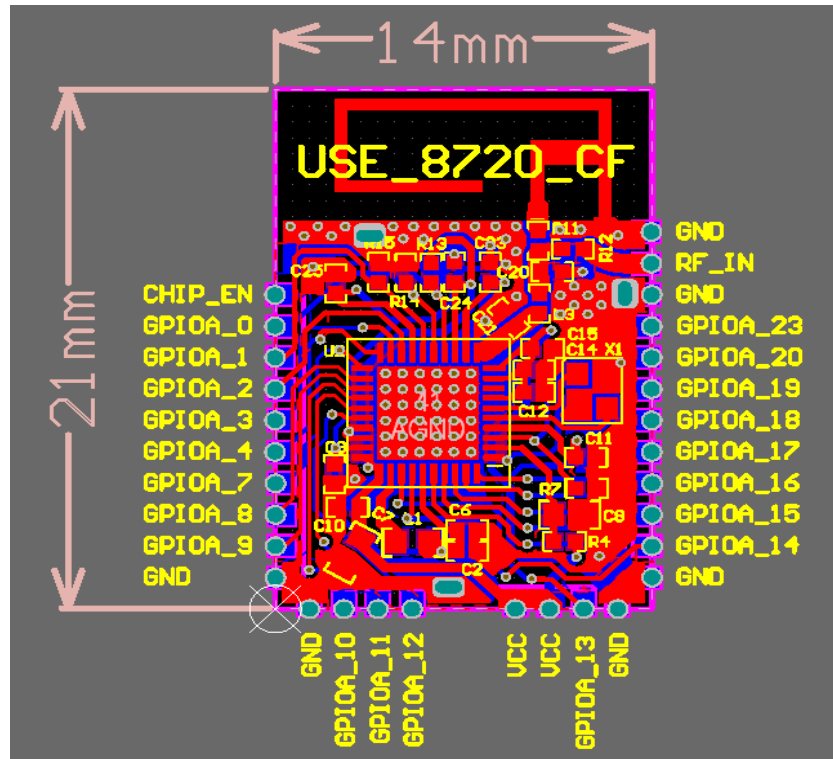


Fig 3(a): Module Pinouts & Dimensions

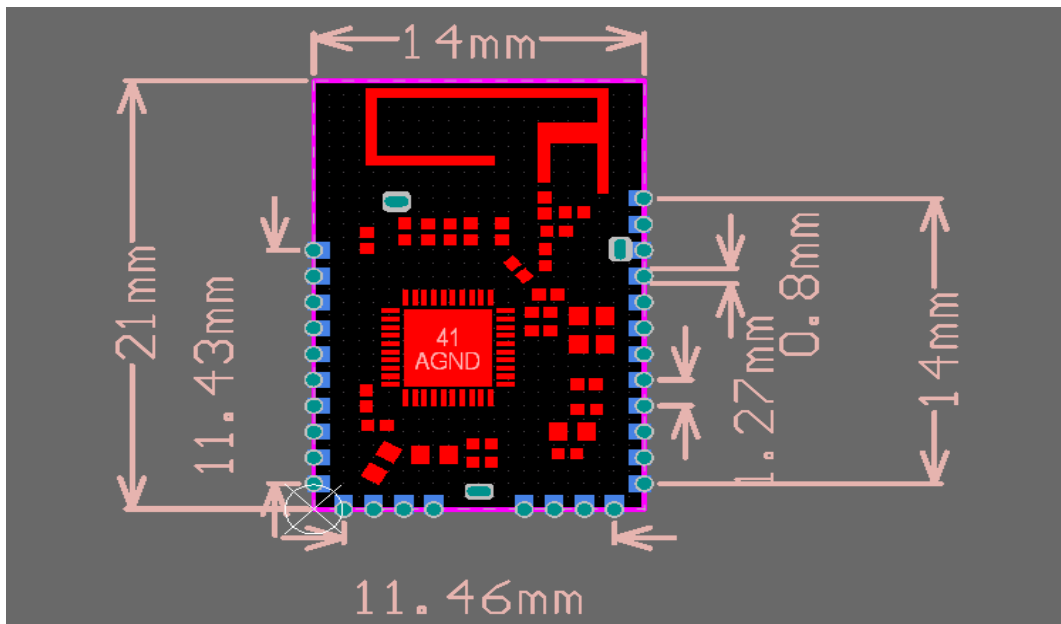


Fig 3(b): Module Pad Dimensions

Recommended Footprint Dimension:

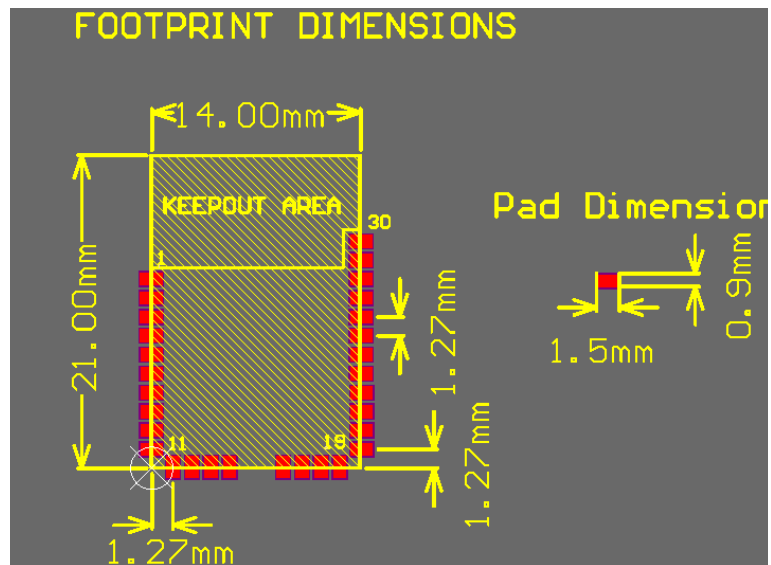


Fig 4: Module Footprint Dimensions

Recommended Footprint clearance:

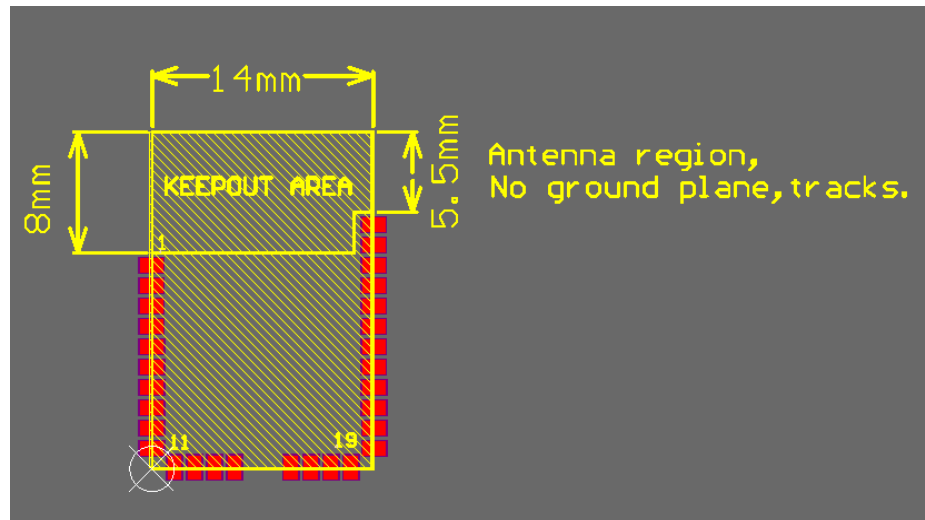


Fig 5: Module Footprint Clearance

Recommended Connections:

The diagram (shown below Fig 6.) of the module application on external filter parts, when needed, please design in the mainboard, the parts parameter can depend on the actual need to change. And pay attention to the increase and decrease ramping of supply voltage faster than 0.5v/ms;

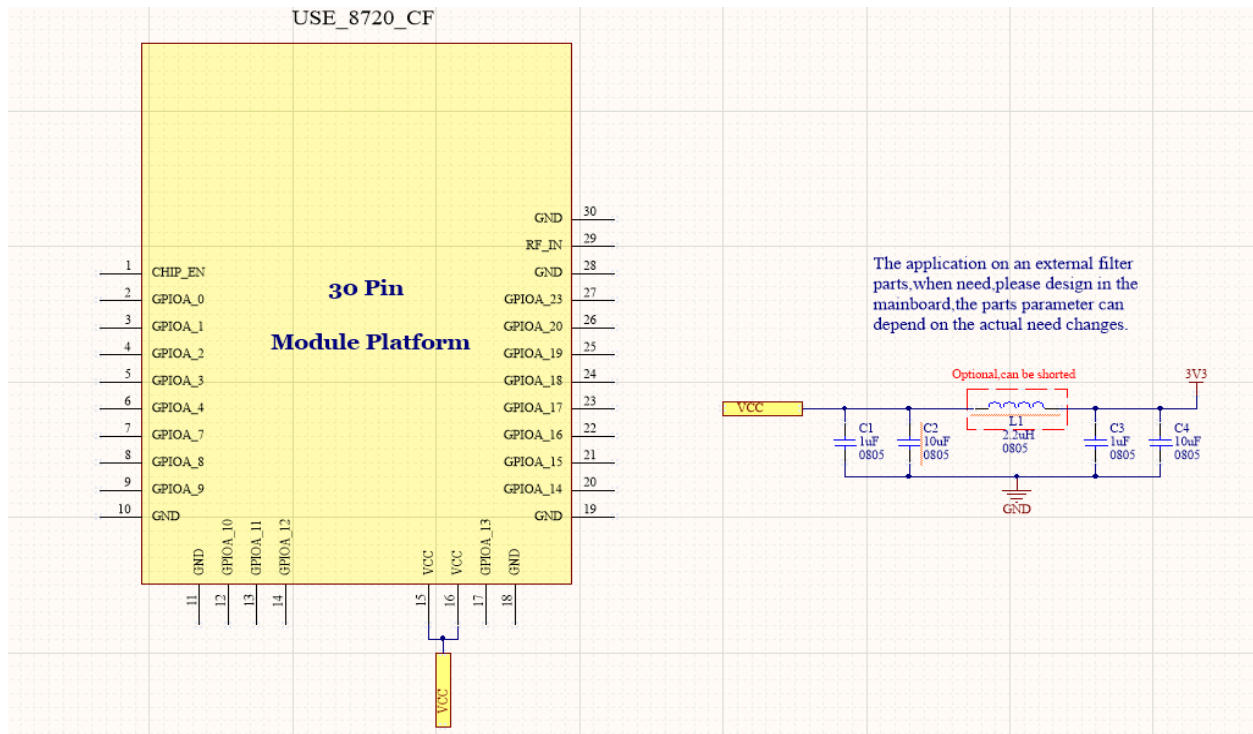


Fig 6

Recommended Assembly:

Signal trace and power supply trace, don't cross layout, as shown in below Fig. To avoid crosstalk, affect the receiving sensitivity.

Assembly recommendation

Underneath the module antenna and RF circuit on the main board PCB copper needs to be cleared, and placed close to the main board edge, as shown in below Fig.

For smaller coin cell batteries, with high worst-case end-of-life equivalent source resistance, a 22- μ F VDDS input capacitor must be used to ensure compliance with this slew rate. Recommended one is to use MLCC for low ESR.

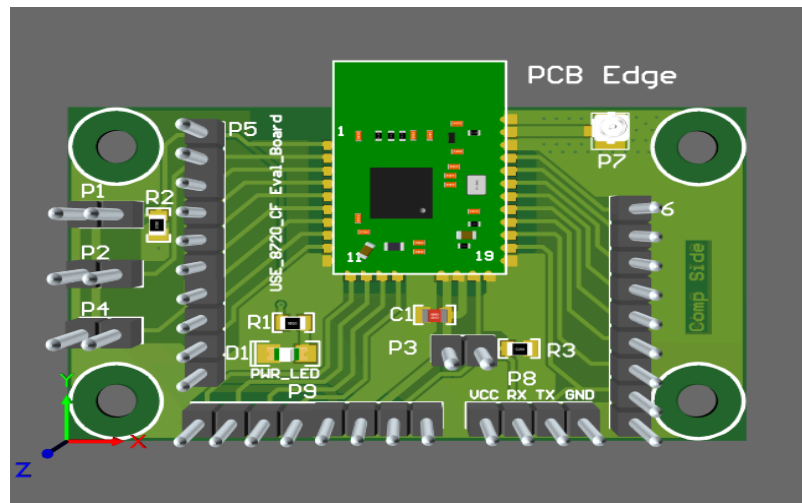


Fig 7(a): Module Assembly 3D

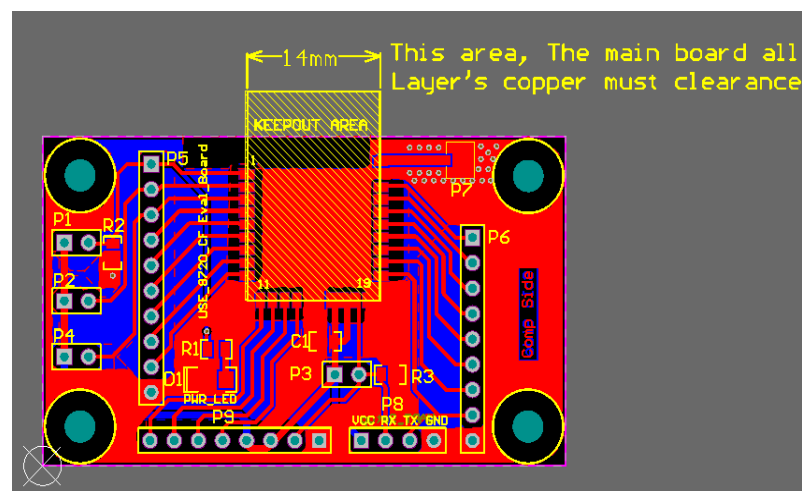


Fig 7(b): Module Assembly 2D

The module assembled in the motherboard, must pay attention to: Copper around the antenna and on all layers must be cleared. The module antenna should be on the edge, and there is no metal or obstruct electromagnetic radiation material to surround the antenna area.

Module placement strategy:

BLE/Sub-1GHz modules are RF devices that require proper placement on PCB to ensure optimal performance. The antenna on the PCB has (about) an omnidirectional radiation pattern. To maximize antenna efficiency, an adequate grounding plane must be provided under the module. However, the areas underneath and surrounding the antenna area must be free of copper. The position of the module on the host board and overall design of the product enclosure contribute to antenna performance. Poor design affects radiation patterns and can result in reflection, diffraction, and/or scattering of the transmitted signal thus limiting the range.

Basic guidelines:

- ❖ Never place the ground plane or route copper traces directly underneath the antenna portion of the module.
- ❖ Never place the antenna close to metallic objects.
- ❖ Keep wiring, components and objects away from the antenna.
- ❖ Do not place the antenna in a metallic or metalized plastic enclosure.
- ❖ Enclosure walls should be 1cm or more away from the antenna in all directions.
- ❖ If possible, mount the antenna overhanging the edge of the host board (Motherboard). Add an uninterrupted ground plane on the host board, directly underneath the top shielded zone of the module, up to the PCB edge. Adding a ground plane will allow traces to be run on the bottom side of the host board if required.
- ❖ If the antenna cannot be mounted in an overhanging position, then provisions must be made to keep the area clear of copper.

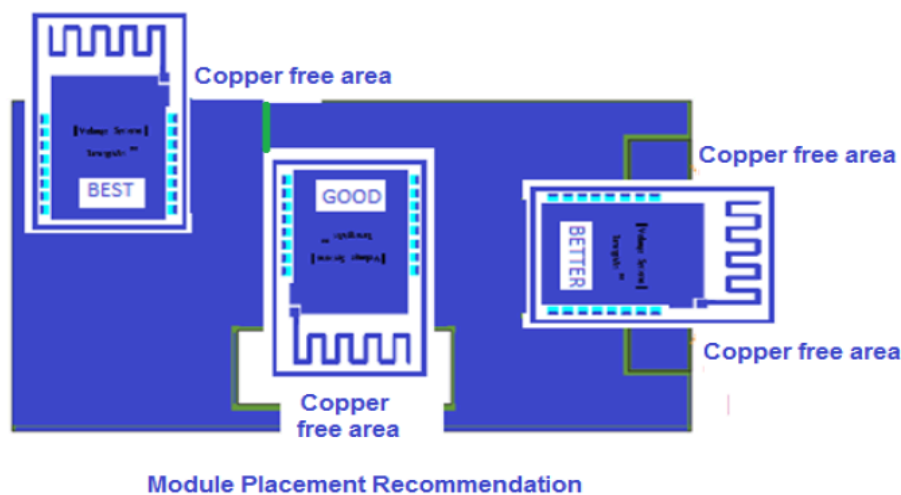
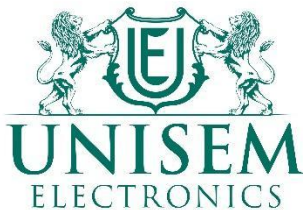


Fig 8

For ordering USE_8720_CF please contact us on info@unisemindia.com.



Disclaimer and Copyright Notice

Unisem India Pvt Ltd or third parties may hold intellectual property rights in the products, names, logos and designs included in this document. Copying, reproduction, or modification of this document or any part thereof is only permitted with the express written permission of Unisem India Pvt Ltd. Disclosure to third parties is permitted for clearly public documents only. The information contained herein is provided “as is”. No warranty of any kind, either express or implied, is made in relation to the accuracy, reliability, fitness for a particular purpose or content of this document. This document may be revised by Unisem India Pvt Ltd at any time. For most recent documents, please visit www.unisemholding.com.